4.2 Local Precompilation

There are several reasons why you might want to precompile your assets locally. Among them are:

- You may not have write access to your production file system.
- You may be deploying to more than one server, and want to avoid duplication of work.
- You may be doing frequent deploys that do not include asset changes.

Local compilation allows you to commit the compiled files into source control, and deploy as normal.

There are three caveats:

- You must not run the Capistrano deployment task that precompiles assets.
- You must ensure any necessary compressors or minifiers are available on your development system.
- You must change the following application configuration setting:

In `config/environments/development.rb`, place the following line:

```
config.assets.prefix = "/dev-assets"
```

The `prefix` change makes Sprockets use a different URL for serving assets in development mode, and pass all requests to Sprockets. The prefix is still set to `/assets` in the production environment. Without this change, the application would serve the precompiled assets from `/assets` in development, and you would not see any local changes until you compile assets again.

In practice, this will allow you to precompile locally, have those files in your working tree, and commit those files to source control when needed. Development mode will work as expected.

4.3 Live Compilation

In some circumstances you may wish to use live compilation. In this mode all requests for assets in the pipeline are handled by Sprockets directly.

To enable this option set:

```
config.assets.compile = true
```

On the first request the assets are compiled and cached as outlined in development above, and the manifest names used in the helpers are altered to include the MD5 hash.

Sprockets also sets the `Cache-Control` HTTP header to `max-age=31536000`. This signals all caches between your server and the client browser that this content (the file served) can be cached for 1 year. The effect of this is to reduce the number of requests for this asset from your server; the asset has a good chance of being in the local browser cache or some intermediate cache.

This mode uses more memory, performs more poorly than the default and is not recommended.

If you are deploying a production application to a system without any pre-existing JavaScript runtimes, you may want to add one to your Gemfile:

```
group :production do
  gem 'therubyracer'
end
```

4.4 CDNs